Chapter 6

Magnetic Disk:

Aluminium Alloy / Glass → substrate

uniformly magnetized coat.

read & write

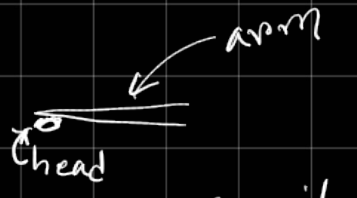
prefer → offer smaller heights.

↑ platter / surface

no. of disks / platters.



Arms



↳ read & write.

Stiffness is required.

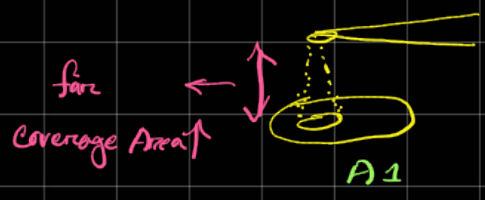
Dynamic inertia



→ inlined?

fly height ↑

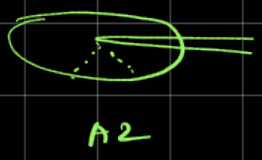
capacity ↓



larger → sector size ↑

↙ read & write on less single sector.

sector accomodate



A1 > A2.

fly height ↓

Smaller → capacity more, data storage more.

↓

sectors size ↓ more sectors accomodate.

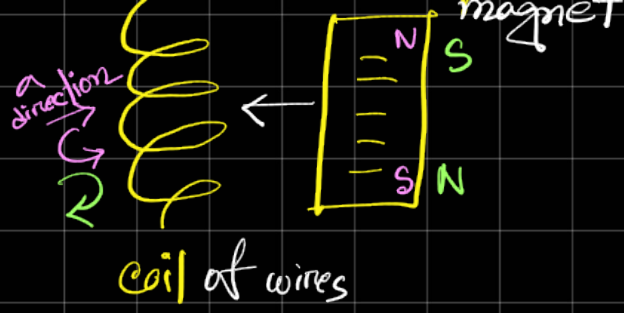
↓

1 packet / group of data.

r/w head separate / together

↓ current

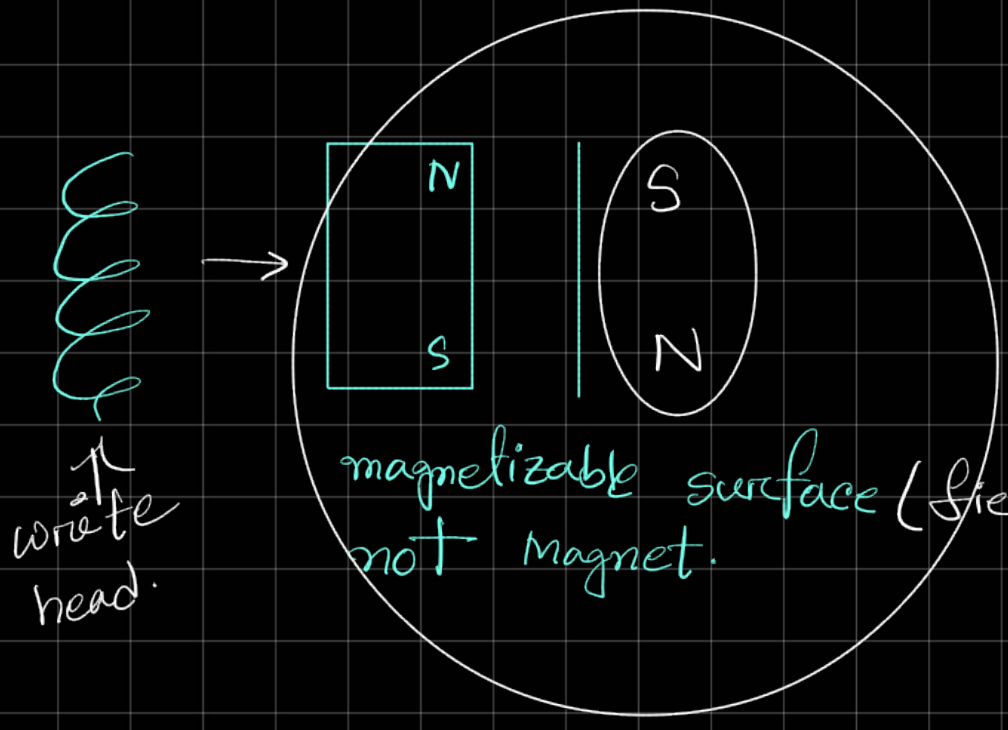
disk surface is magnet



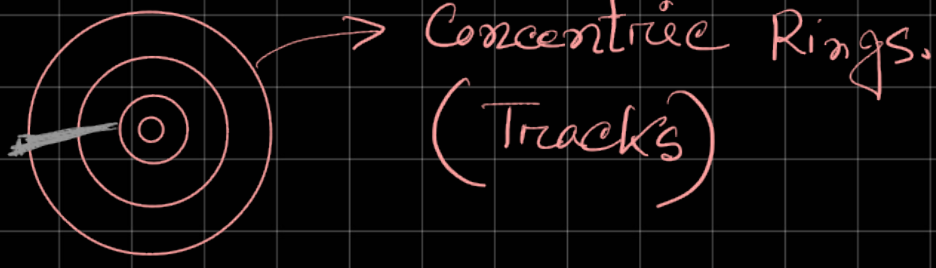
Read operation



Write operation



Disk data layout:

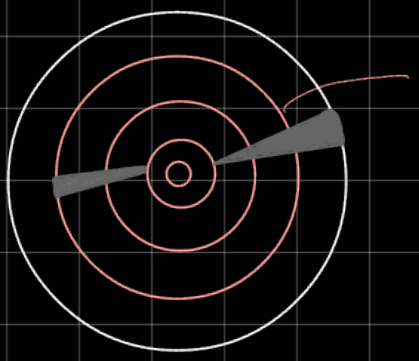


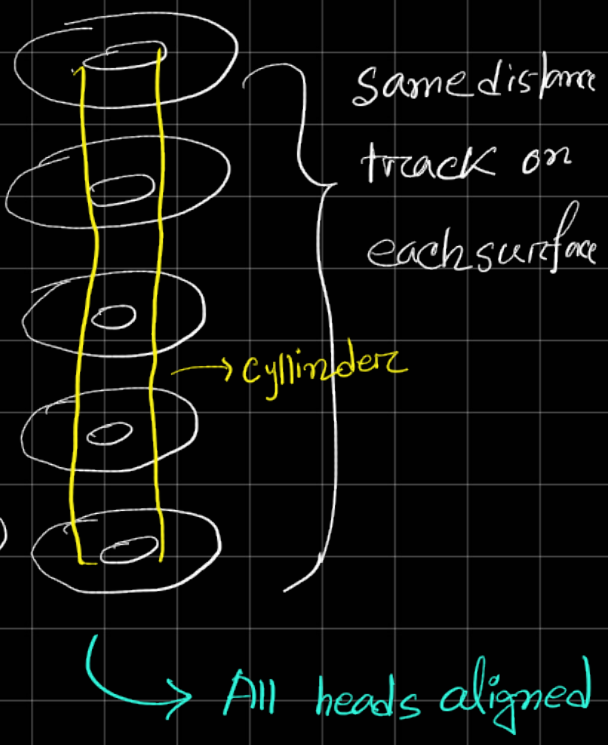
Rltnship among tracks

closer to center → track size smaller.

same for sectors.

Greay lines → Inter sector / track gap.





how many tracks are possible?

$5 ? \times 2 = 10$ ✓

Why consider all surfaces at a time?

10 bit data → 5 Platters → 10 surfaces.